

		Isopropanol	Water	DiMth Sulfoxide	Bottom Tempera	Vapour IPA	Vapour WT	Vapour DMSO
Time (h)	Reflux Ratio	kmol/h	kmol/h	kmol/h	C	kmol/h	kmol/h	kmol/h
0	0	0.000835523	0.000412389	0	80.24	0.000835523	0.000412389	0.00E+00
0.05	4	4.74825	0.00288502	5.25E-05	80.3893	23.74125	0.0144251	2.63E-04
0.1	4	4.74609	0.00284451	5.26E-05	80.5409	23.73045	0.01422255	2.63E-04
0.15	4	4.74349	0.00280624	5.26E-05	80.6946	23.71745	0.0140312	2.63E-04
0.2	4	4.74086	0.00276982	5.26E-05	80.8503	23.7043	0.0138491	2.63E-04
0.25	4	4.73856	0.00273532	5.26E-05	81.008	23.6928	0.0136766	2.63E-04
0.3	4	4.73628	0.0027014	5.26E-05	81.1676	23.6814	0.013507	2.63E-04
0.35	4	4.73417	0.00266993	5.26E-05	81.329	23.67085	0.01334965	2.63E-04
0.4	4	4.73199	0.0026383	5.26E-05	81.4922	23.65995	0.0131915	2.63E-04
0.45	4	4.73005	0.00260865	5.26E-05	81.657	23.65025	0.01304325	2.63E-04
0.5	4	4.72789	0.00257872	5.26E-05	81.8234	23.63945	0.0128936	2.63E-04
0.55	4	4.72593	0.00255025	5.26E-05	81.9915	23.62965	0.01275125	2.63E-04
0.6	4	4.724	0.00252264	5.26E-05	82.161	23.62	0.0126132	2.63E-04
0.65	4	4.72199	0.00249535	5.26E-05	82.332	23.60995	0.01247675	2.63E-04
0.7	4	4.72026	0.00246996	5.26E-05	82.5045	23.6013	0.0123498	2.63E-04
0.75	4	4.71817	0.00244314	5.26E-05	82.6784	23.59085	0.0122157	2.63E-04
0.8	4	4.71621	0.00241803	5.26E-05	82.8536	23.58105	0.01209015	2.63E-04
0.85	4	4.71428	0.00239336	5.26E-05	83.0302	23.5714	0.0119668	2.63E-04
0.9	4	4.71296	0.00237209	5.26E-05	83.2081	23.5648	0.01186045	2.63E-04
0.95	4	4.71093	0.00234792	5.26E-05	83.3873	23.55465	0.0117396	2.63E-04
1	4	4.70911	0.00232527	5.26E-05	83.5678	23.54555	0.01162635	2.63E-04
1.05	4	4.70705	0.00230216	5.26E-05	83.7496	23.53525	0.0115108	2.63E-04
1.1	4	4.70549	0.0022817	5.26E-05	83.9326	23.52745	0.0114085	2.63E-04
1.15	4	4.70379	0.00226099	5.25E-05	84.1168	23.51895	0.01130495	2.63E-04
1.2	4	4.70223	0.0022414	5.25E-05	84.3023	23.51115	0.011207	2.63E-04
1.25	4	4.7002	0.00222019	5.25E-05	84.489	23.501	0.01110095	2.63E-04
1.3	4	4.69865	0.00220157	5.25E-05	84.6768	23.49325	0.01100785	2.63E-04
1.35	4	4.69705	0.00218309	5.25E-05	84.8659	23.48525	0.01091545	2.63E-04

1.4	4	4.69511	0.0021636	5.25E-05	85.0562	23.47555	0.010818	2.63E-04
1.45	4	4.6934	0.00214547	5.25E-05	85.2477	23.467	0.01072735	2.63E-04
1.5	4	4.69175	0.00212796	5.25E-05	85.4404	23.45875	0.0106398	2.63E-04
1.55	4	4.69004	0.00211064	5.25E-05	85.6342	23.4502	0.0105532	2.63E-04
1.6	4	4.68822	0.00209314	5.25E-05	85.8293	23.4411	0.0104657	2.63E-04
1.65	4	4.68614	0.00207499	5.25E-05	86.0256	23.4307	0.01037495	2.63E-04
1.7	4	4.68465	0.00205973	5.25E-05	86.2231	23.42325	0.01029865	2.63E-04
1.75	4	4.68272	0.00204299	5.25E-05	86.4218	23.4136	0.01021495	2.63E-04
1.8	4	4.68112	0.00202784	5.25E-05	86.6218	23.4056	0.0101392	2.63E-04
1.85	4	4.67904	0.00201107	5.25E-05	86.8229	23.3952	0.01005535	2.63E-04
1.9	4	4.67717	0.0019956	5.25E-05	87.0254	23.38585	0.009978	2.63E-04
1.95	4	4.67549	0.00198116	5.25E-05	87.229	23.37745	0.0099058	2.63E-04
2	4	4.67357	0.00196614	5.25E-05	87.4339	23.36785	0.0098307	2.63E-04
2.05	4	4.67158	0.00195111	5.25E-05	87.6401	23.3579	0.00975555	2.63E-04
2.1	4	4.66943	0.0019358	5.25E-05	87.8476	23.34715	0.009679	2.63E-04
2.15	4	4.66773	0.00192248	5.25E-05	88.0564	23.33865	0.0096124	2.63E-04
2.2	4	4.6658	0.00190869	5.25E-05	88.2665	23.329	0.00954345	2.63E-04
2.25	4	4.66385	0.00189507	5.25E-05	88.478	23.31925	0.00947535	2.63E-04
2.3	4	4.6617	0.00188093	5.25E-05	88.6908	23.3085	0.00940465	2.63E-04
2.35	4	4.65965	0.00186754	5.25E-05	88.9049	23.29825	0.0093377	2.63E-04
2.4	4	4.65784	0.00185535	5.25E-05	89.1204	23.2892	0.00927675	2.63E-04
2.45	4	4.65556	0.00184159	5.25E-05	89.3373	23.2778	0.00920795	2.63E-04
2.5	4	4.65378	0.00183001	5.25E-05	89.5557	23.2689	0.00915005	2.63E-04
2.55	4	4.65113	0.00181548	5.25E-05	89.7755	23.25565	0.0090774	2.63E-04
2.6	4	4.64922	0.00180395	5.25E-05	89.9968	23.2461	0.00901975	2.63E-04
2.65	4	4.6469	0.00179112	5.25E-05	90.2195	23.2345	0.0089556	2.63E-04
2.7	4	4.64473	0.00177924	5.25E-05	90.4437	23.22365	0.0088962	2.63E-04
2.75	4	4.64254	0.00176748	5.25E-05	90.6695	23.2127	0.0088374	2.63E-04
2.8	4	4.64024	0.00175558	5.25E-05	90.8968	23.2012	0.0087779	2.63E-04
2.85	4	4.63805	0.00174424	5.25E-05	91.1257	23.19025	0.0087212	2.63E-04
2.9	4	4.63557	0.00173218	5.25E-05	91.3562	23.17785	0.0086609	2.63E-04
2.95	4	4.63314	0.00172065	5.25E-05	91.5882	23.1657	0.00860325	2.63E-04
3	4	4.63074	0.00170934	5.25E-05	91.822	23.1537	0.0085467	2.63E-04
3.05	4	4.62833	0.00169829	5.25E-05	92.0574	23.14165	0.00849145	2.63E-04

3.1	4	4.62548	0.00168603	5.25E-05	92.2945	23.1274	0.00843015	2.63E-04
3.15	4	4.62324	0.00167595	5.25E-05	92.5333	23.1162	0.00837975	2.63E-04
3.2	4	4.62039	0.00166416	5.25E-05	92.7739	23.10195	0.0083208	2.63E-04
3.25	4	4.61793	0.00165392	5.25E-05	93.0162	23.08965	0.0082696	2.62E-04
3.3	4	4.6152	0.00164289	5.25E-05	93.2604	23.076	0.00821445	2.62E-04
3.35	4	4.61251	0.0016323	5.25E-05	93.5064	23.06255	0.0081615	2.62E-04
3.4	4	4.60976	0.00162173	5.25E-05	93.7543	23.0488	0.00810865	2.62E-04
3.45	4	4.60729	0.00161231	5.25E-05	94.0041	23.03645	0.00806155	2.62E-04
3.5	4	4.60432	0.00160146	5.25E-05	94.2558	23.0216	0.0080073	2.62E-04
3.55	4	4.60138	0.00159093	5.25E-05	94.5094	23.0069	0.00795465	2.62E-04
3.6	4	4.59837	0.00158041	5.25E-05	94.7651	22.99185	0.00790205	2.62E-04
3.65	4	4.5955	0.00157053	5.25E-05	95.0228	22.9775	0.00785265	2.62E-04
3.7	4	4.59246	0.00156035	5.25E-05	95.2825	22.9623	0.00780175	2.62E-04
3.75	4	4.58941	0.00155037	5.25E-05	95.5444	22.94705	0.00775185	2.62E-04
3.8	4	4.58637	0.0015406	5.25E-05	95.8084	22.93185	0.007703	2.62E-04
3.85	4	4.5831	0.00153037	5.25E-05	96.0745	22.9155	0.00765185	2.62E-04
3.9	4	4.57989	0.00152043	5.25E-05	96.3429	22.89945	0.00760215	2.62E-04
3.95	4	4.5765	0.00151026	5.25E-05	96.6135	22.8825	0.0075513	2.62E-04
4	4	4.57323	0.00150054	5.25E-05	96.8864	22.86615	0.0075027	2.62E-04
4.05	4	4.56992	0.00149105	5.25E-05	97.1616	22.8496	0.00745525	2.62E-04
4.1	4	4.5664	0.00148111	5.25E-05	97.4392	22.832	0.00740555	2.62E-04
4.15	4	4.56307	0.00147178	5.24E-05	97.7191	22.81535	0.0073589	2.62E-04
4.2	4	4.55951	0.00146215	5.24E-05	98.0016	22.79755	0.00731075	2.62E-04
4.25	4	4.55559	0.00145155	5.24E-05	98.2865	22.77795	0.00725775	2.62E-04
4.3	4	4.55195	0.00144201	5.24E-05	98.5739	22.75975	0.00721005	2.62E-04
4.35	4	4.54813	0.00143219	5.24E-05	98.8639	22.74065	0.00716095	2.62E-04
4.4	4	4.54448	0.00142301	5.24E-05	99.1566	22.7224	0.00711505	2.62E-04
4.45	4	4.54032	0.00141255	5.24E-05	99.4518	22.7016	0.00706275	2.62E-04
4.5	4	4.53641	0.00140312	5.24E-05	99.7498	22.68205	0.0070156	2.62E-04
4.55	4	4.53232	0.00139328	5.24E-05	100.051	22.6616	0.0069664	2.62E-04
4.6	4	4.52857	0.00138461	5.24E-05	100.354	22.64285	0.00692305	2.62E-04
4.65	4	4.52415	0.00137429	5.24E-05	100.661	22.62075	0.00687145	2.62E-04
4.7	4	4.52	0.00136485	5.24E-05	100.97	22.6	0.00682425	2.62E-04
4.75	4	4.51578	0.00135549	5.24E-05	101.282	22.5789	0.00677745	2.62E-04

4.8	4	4.51148	0.0013461	5.24E-05	101.598	22.5574	0.0067305	2.62E-04
4.85	4	4.5067	0.00133553	5.24E-05	101.916	22.5335	0.00667765	2.62E-04
4.9	4	4.50223	0.00132605	5.24E-05	102.237	22.51115	0.00663025	2.62E-04
4.95	4	4.49752	0.00131608	5.24E-05	102.562	22.4876	0.0065804	2.62E-04
5	4	4.49281	0.00130638	5.24E-05	102.89	22.46405	0.0065319	2.62E-04
5.05	4	4.4878	0.00129607	5.24E-05	103.221	22.439	0.00648035	2.62E-04
5.1	4	4.48312	0.00128682	5.24E-05	103.556	22.4156	0.0064341	2.62E-04
5.15	4	4.4782	0.00127709	5.24E-05	103.894	22.391	0.00638545	2.62E-04
5.2	4	4.47315	0.00126726	5.24E-05	104.235	22.36575	0.0063363	2.62E-04
5.25	4	4.46758	0.00125627	5.23E-05	104.58	22.3379	0.00628135	2.62E-04
5.3	4	4.46232	0.00124632	5.23E-05	104.929	22.3116	0.0062316	2.62E-04
5.35	4	4.4569	0.00123614	5.23E-05	105.281	22.2845	0.0061807	2.62E-04
5.4	4	4.45142	0.00122604	5.23E-05	105.637	22.2571	0.0061302	2.62E-04
5.45	4	4.44576	0.00121562	5.23E-05	105.997	22.2288	0.0060781	2.62E-04
5.5	4	4.43994	0.00120509	5.23E-05	106.361	22.1997	0.00602545	2.62E-04
5.55	4	4.43417	0.00119491	5.23E-05	106.728	22.17085	0.00597455	2.62E-04
5.6	4	4.42853	0.00118519	5.23E-05	107.1	22.14265	0.00592595	2.62E-04
5.65	4	4.42221	0.00117398	5.23E-05	107.476	22.11105	0.0058699	2.61E-04
5.7	4	4.41623	0.00116395	5.23E-05	107.856	22.08115	0.00581975	2.61E-04
5.75	4	4.40982	0.00115304	5.23E-05	108.24	22.0491	0.0057652	2.61E-04
5.8	4	4.40335	0.0011422	5.23E-05	108.629	22.01675	0.005711	2.61E-04
5.85	4	4.39659	0.00113084	5.23E-05	109.022	21.98295	0.0056542	2.61E-04
5.9	4	4.39007	0.0011204	5.23E-05	109.419	21.95035	0.005602	2.61E-04
5.95	4	4.38297	0.00110872	5.23E-05	109.821	21.91485	0.0055436	2.61E-04
6	4	4.37611	0.00109786	5.22E-05	110.228	21.88055	0.0054893	2.61E-04
6.05	4	4.36914	0.00108701	5.22E-05	110.64	21.8457	0.00543505	2.61E-04
6.1	4	4.36169	0.00107519	5.22E-05	111.056	21.80845	0.00537595	2.61E-04
6.15	4	4.35422	0.00106365	5.22E-05	111.477	21.7711	0.00531825	2.61E-04
6.2	4	4.34668	0.00105224	5.22E-05	111.904	21.7334	0.0052612	2.61E-04
6.25	4	4.33895	0.00104061	5.22E-05	112.335	21.69475	0.00520305	2.61E-04
6.3	4	4.33119	0.00102924	5.22E-05	112.772	21.65595	0.0051462	2.61E-04
6.35	4	4.32343	0.00101799	5.22E-05	113.214	21.61715	0.00508995	2.61E-04
6.4	4	4.315	0.00100563	5.22E-05	113.661	21.575	0.00502815	2.61E-04
6.45	4	4.30664	0.000993692	5.22E-05	114.115	21.5332	0.00496846	2.61E-04

6.5	4	4.29806	0.000981575	5.22E-05	114.573	21.4903	0.004907875	2.61E-04
6.55	4	4.28916	0.000968996	5.21E-05	115.038	21.4458	0.00484498	2.61E-04
6.6	4	4.2804	0.000957082	5.21E-05	115.508	21.402	0.00478541	2.61E-04
6.65	4	4.2711	0.000944237	5.21E-05	115.985	21.3555	0.004721185	2.61E-04
6.7	4	4.26198	0.000932146	5.21E-05	116.467	21.3099	0.00466073	2.61E-04
6.75	4	4.25245	0.000919521	5.21E-05	116.956	21.26225	0.004597605	2.61E-04
6.8	4	4.24232	0.000905972	5.21E-05	117.451	21.2116	0.00452986	2.60E-04
6.85	4	4.23272	0.000893803	5.21E-05	117.952	21.1636	0.004469015	2.60E-04
6.9	4	4.22228	0.000880313	5.21E-05	118.46	21.1114	0.004401565	2.60E-04
6.95	4	4.21201	0.000867467	5.21E-05	118.975	21.06005	0.004337335	2.60E-04
7	4	4.20137	0.000854331	5.20E-05	119.497	21.00685	0.004271655	2.60E-04
7.05	4	4.19058	0.00084131	5.20E-05	120.026	20.9529	0.00420655	2.60E-04
7.1	4	4.17958	0.000828147	5.20E-05	120.561	20.8979	0.004140735	2.60E-04
7.15	4	4.16822	0.000814788	5.20E-05	121.104	20.8411	0.00407394	2.60E-04
7.2	4	4.15673	0.000801459	5.20E-05	121.655	20.78365	0.004007295	2.60E-04
7.25	4	4.14485	0.00078789	5.20E-05	122.213	20.72425	0.00393945	2.60E-04
7.3	4	4.13301	0.000774939	5.20E-05	122.778	20.66505	0.003874695	2.60E-04
7.35	4	4.12065	0.000761357	5.19E-05	123.352	20.60325	0.003806785	2.60E-04
7.4	4	4.10794	0.000747647	5.19E-05	123.933	20.5397	0.003738235	2.60E-04
7.45	4	4.09522	0.00073438	5.19E-05	124.523	20.4761	0.0036719	2.60E-04
7.5	4	4.08194	0.000720657	5.19E-05	125.12	20.4097	0.003603285	2.59E-04
7.55	4	4.06858	0.000707225	5.19E-05	125.727	20.3429	0.003536125	2.59E-04
7.6	4	4.05493	0.000693863	5.19E-05	126.341	20.27465	0.003469315	2.59E-04
7.65	4	4.04065	0.000680128	5.18E-05	126.965	20.20325	0.00340064	2.59E-04
7.7	4	4.02614	0.000666456	5.18E-05	127.597	20.1307	0.00333228	2.59E-04
7.75	4	4.01157	0.000653377	5.18E-05	128.238	20.05785	0.003266885	2.59E-04
7.8	4	3.99702	0.000640752	5.18E-05	128.889	19.9851	0.00320376	2.59E-04
7.85	4	3.98125	0.000627097	5.18E-05	129.548	19.90625	0.003135485	2.59E-04
7.9	4	3.96602	0.000614797	5.17E-05	130.218	19.8301	0.003073985	2.59E-04
7.95	4	3.95022	0.000602191	5.17E-05	130.897	19.7511	0.003010955	2.59E-04
8	4	3.93393	0.00058987	5.17E-05	131.585	19.66965	0.00294935	2.58E-04
8.05	4	3.91765	0.000578075	5.17E-05	132.284	19.58825	0.002890375	2.58E-04
8.1	4	3.90052	0.000565998	5.16E-05	132.993	19.5026	0.00282999	2.58E-04
8.15	4	3.88355	0.00055493	5.16E-05	133.712	19.41775	0.00277465	2.58E-04

8.2	4	3.86559	0.000543577	5.16E-05	134.441	19.32795	0.002717885	2.58E-04
8.25	4	3.84773	0.000533154	5.16E-05	135.181	19.23865	0.00266577	2.58E-04
8.3	4	3.82936	0.000523233	5.15E-05	135.932	19.1468	0.002616165	2.58E-04
8.35	4	3.81082	0.000514212	5.15E-05	136.693	19.0541	0.00257106	2.58E-04
8.4	4	3.792	0.000506069	5.15E-05	137.466	18.96	0.002530345	2.57E-04
8.45	4	3.77217	0.000498134	5.14E-05	138.249	18.86085	0.00249067	2.57E-04
8.5	4	3.75316	0.000492822	5.14E-05	139.044	18.7658	0.00246411	2.57E-04
8.55	4	3.73312	0.000488285	5.14E-05	139.85	18.6656	0.002441425	2.57E-04
8.6	4	3.71261	0.000485572	5.14E-05	140.668	18.56305	0.00242786	2.57E-04
8.65	4	3.69179	0.000485592	5.13E-05	141.497	18.45895	0.00242796	2.57E-04
8.7	4	3.67112	0.000490071	5.13E-05	142.338	18.3556	0.002450355	2.56E-04
8.75	4	3.64946	0.000499763	5.12E-05	143.191	18.2473	0.002498815	2.56E-04
8.8	4	3.62776	0.000520207	5.12E-05	144.055	18.1388	0.002601035	2.56E-04
8.85	4	3.60578	0.000562569	5.12E-05	144.932	18.0289	0.002812845	2.56E-04
8.9	4	3.58342	0.000673605	5.11E-05	145.82	17.9171	0.003368025	2.56E-04
8.95	4	3.51617	0.0498227	4.62E-05	146.713	17.58085	0.2491135	2.31E-04
9	4	3.26039	0.310652	2.57E-05	147.575	16.30195	1.55326	1.28E-04
9.05	4	3.01136	0.571136	1.28E-05	148.407	15.0568	2.85568	6.42E-05
9.1	5	2.42163	1.19538	2.90E-14	149.014	14.52978	7.17228	1.74E-13
9.15	5	2.41565	1.19244	0	149.631	14.4939	7.15464	0.00E+00
9.2	5	2.40936	1.18937	0	150.259	14.45616	7.13622	0.00E+00
9.25	5	2.28667	1.31811	0	150.875	13.72002	7.90866	0.00E+00
9.3	5	2.05798	1.56567	0	151.456	12.34788	9.39402	0.00E+00
9.35	5	1.84321	1.79125	0	152.003	11.05926	10.7475	0.00E+00
9.4	5	1.64351	2.00183	1.00E-15	152.517	9.86106	12.01098	6.00E-15
9.45	5	1.45888	2.19885	1.00E-15	153	8.75328	13.1931	6.00E-15
9.5	5	1.28953	2.38184	1.00E-15	153.453	7.73718	14.29104	6.00E-15
9.55	5	1.13503	2.54986	2.00E-15	153.879	6.81018	15.29916	1.20E-14
9.6	5	0.995232	2.70335	2.00E-15	154.279	5.971392	16.2201	1.20E-14
9.65	5	0.869561	2.84279	3.00E-15	154.656	5.217366	17.05674	1.80E-14
9.7	5	0.75709	2.96817	3.00E-15	155.012	4.54254	17.80902	1.80E-14
9.75	5	0.657082	3.08063	4.00E-15	155.35	3.942492	18.48378	2.40E-14
9.8	5	0.568617	3.18106	4.00E-15	155.671	3.411702	19.08636	2.40E-14
9.85	5	0.490776	3.27082	5.00E-15	155.977	2.944656	19.62492	3.00E-14

9.9	5	0.422482	3.35007	5.00E-15	156.27	2.534892	20.10042	3.00E-14
9.95	5	0.362908	3.42076	5.00E-15	156.552	2.177448	20.52456	3.00E-14
10	5	0.311072	3.48331	6.00E-15	156.825	1.866432	20.89986	3.60E-14
10.05	5	0.266129	3.53872	6.00E-15	157.089	1.596774	21.23232	3.60E-14
10.1	5	0.227259	3.58737	6.00E-15	157.347	1.363554	21.52422	3.60E-14
10.15	5	0.193761	3.63051	6.00E-15	157.599	1.162566	21.78306	3.60E-14
10.2	5	0.164942	3.6683	7.00E-15	157.847	0.989652	22.0098	4.20E-14
10.25	5	0.140217	3.70159	7.00E-15	158.091	0.841302	22.20954	4.20E-14
10.3	5	0.119049	3.73102	7.00E-15	158.331	0.714294	22.38612	4.20E-14
10.35	5	0.100947	3.75655	7.00E-15	158.57	0.605682	22.5393	4.20E-14
10.4	5	0.0854974	3.77874	7.00E-15	158.807	0.5129844	22.67244	4.20E-14
10.45	5	0.0723335	3.79807	7.00E-15	159.043	0.434001	22.78842	4.20E-14
10.5	5	0.0611276	3.8144	7.00E-15	159.278	0.3667656	22.8864	4.20E-14
10.55	5	0.0516095	3.82869	7.00E-15	159.512	0.309657	22.97214	4.20E-14
10.6	5	0.0435304	3.84079	8.00E-15	159.747	0.2611824	23.04474	4.80E-14
10.65	5	0.0366803	3.85099	8.00E-15	159.982	0.2200818	23.10594	4.80E-14
10.7	5	0.03088	3.85957	8.00E-15	160.217	0.18528	23.15742	4.80E-14
10.75	5	0.025974	3.8668	8.00E-15	160.453	0.155844	23.2008	4.80E-14
10.8	5	0.0218284	3.87282	8.00E-15	160.69	0.1309704	23.23692	4.80E-14
10.85	4	0.0182999	4.61991	1.10E-14	160.972	0.0914995	23.09955	5.50E-14
10.9	4	0.0153682	4.62298	1.10E-14	161.256	0.076841	23.1149	5.50E-14
10.95	4	0.0128925	4.62487	1.10E-14	161.543	0.0644625	23.12435	5.50E-14
11	4	0.0108061	4.62666	1.10E-14	161.831	0.0540305	23.1333	5.50E-14
11.05	4	0.00904742	4.62735	1.10E-14	162.122	0.0452371	23.13675	5.50E-14
11.1	4	0.00756724	4.62739	1.10E-14	162.414	0.0378362	23.13695	5.50E-14
11.15	4	0.00632327	4.62727	1.10E-14	162.71	0.03161635	23.13635	5.50E-14
11.2	4	0.00527839	4.62675	1.10E-14	163.007	0.02639195	23.13375	5.50E-14
11.25	4	0.00440161	4.62574	1.10E-14	163.308	0.02200805	23.1287	5.50E-14
11.3	4	0.00366674	4.62455	1.10E-14	163.61	0.0183337	23.12275	5.50E-14
11.35	4	0.00305118	4.6227	1.10E-14	163.916	0.0152559	23.1135	5.50E-14
11.4	4	0.00253648	4.62098	1.10E-14	164.224	0.0126824	23.1049	5.50E-14
11.45	4	0.00210643	4.61924	1.10E-14	164.535	0.01053215	23.0962	5.50E-14
11.5	4	0.00174728	4.61692	1.10E-14	164.849	0.0087364	23.0846	5.50E-14
11.55	4	0.0014479	4.61482	1.10E-14	165.166	0.0072395	23.0741	5.50E-14

11.6	4	0.0011984	4.61219	1.10E-14	165.485	0.005992	23.06095	5.50E-14
11.65	4	0.000990835	4.60968	1.10E-14	165.808	0.004954175	23.0484	5.50E-14
11.7	4	0.000818307	4.60714	1.10E-14	166.134	0.004091535	23.0357	5.50E-14
11.75	4	0.000675026	4.60435	1.10E-14	166.462	0.00337513	23.02175	5.50E-14
11.8	4	0.00055617	4.60137	1.10E-14	166.794	0.00278085	23.00685	5.50E-14
11.85	4	0.000457717	4.59846	1.10E-14	167.129	0.002288585	22.9923	5.50E-14
11.9	4	0.000376243	4.59553	1.10E-14	167.468	0.001881215	22.97765	5.50E-14
11.95	4	0.000308865	4.59202	1.10E-14	167.81	0.001544325	22.9601	5.50E-14
12	4	0.000253272	4.58908	1.10E-14	168.155	0.00126636	22.9454	5.50E-14
12.05	4	0.000207409	4.58577	1.10E-14	168.503	0.001037045	22.92885	5.50E-14
12.1	4	0.000169634	4.58236	1.10E-14	168.855	0.00084817	22.9118	5.50E-14
12.15	4	0.000138559	4.57899	1.10E-14	169.211	0.000692795	22.89495	5.50E-14
12.2	4	0.000113025	4.5755	1.10E-14	169.57	0.000565125	22.8775	5.50E-14
12.25	4	9.21E-05	4.57195	1.10E-14	169.932	0.000460361	22.85975	5.50E-14
12.3	4	7.49E-05	4.56805	1.10E-14	170.299	0.000374481	22.84025	5.50E-14
12.35	4	6.08E-05	4.56459	1.10E-14	170.669	0.000304225	22.82295	5.50E-14
12.4	4	4.94E-05	4.56072	1.10E-14	171.043	0.000246774	22.8036	5.50E-14
12.45	4	4.00E-05	4.55683	1.10E-14	171.421	0.000199885	22.78415	5.50E-14
12.5	4	3.23E-05	4.55291	1.10E-14	171.803	0.000161663	22.76455	5.50E-14
12.55	4	2.61E-05	4.54914	1.10E-14	172.188	0.000130556	22.7457	5.50E-14
12.6	4	2.11E-05	4.54519	1.10E-14	172.578	0.000105266	22.72595	5.50E-14
12.65	4	1.69E-05	4.54073	1.10E-14	172.972	8.47335E-05	22.70365	5.50E-14
12.7	4	1.36E-05	4.5366	1.10E-14	173.37	6.80995E-05	22.683	5.50E-14
12.75	4	1.09E-05	4.53217	1.10E-14	173.773	0.000054638	22.66085	5.50E-14
12.8	4	8.75E-06	4.52781	1.10E-14	174.179	4.37634E-05	22.63905	5.50E-14
12.85	4	7.00E-06	4.52375	1.10E-14	174.591	3.49938E-05	22.61875	5.50E-14
12.9	4	5.59E-06	4.51899	1.10E-14	175.006	2.79275E-05	22.59495	5.50E-14
12.95	4	4.45E-06	4.51464	1.10E-14	175.426	2.22492E-05	22.5732	5.50E-14
13	4	3.54E-06	4.50984	1.10E-14	175.851	1.76903E-05	22.5492	5.50E-14
13.05	4	2.81E-06	4.50525	1.10E-14	176.28	1.40388E-05	22.52625	5.50E-14
13.1	4	2.22E-06	4.5003	1.10E-14	176.715	1.1118E-05	22.5015	5.50E-14
13.15	4	1.76E-06	4.49526	1.10E-14	177.153	8.7865E-06	22.4763	5.50E-14
13.2	4	1.39E-06	4.49028	1.10E-14	177.597	6.9293E-06	22.4514	5.50E-14
13.25	4	1.09E-06	4.48517	1.10E-14	178.046	5.45245E-06	22.42585	5.50E-14

13.3	4	8.56E-07	4.47999	1.10E-14	178.5	4.28054E-06	22.39995	5.50E-14
13.35	4	6.71E-07	4.47476	1.10E-14	178.959	3.35253E-06	22.3738	5.50E-14
13.4	4	5.24E-07	4.46941	1.10E-14	179.423	2.61919E-06	22.34705	5.50E-14
13.45	4	4.08E-07	4.46391	1.10E-14	179.893	2.04099E-06	22.31955	5.50E-14
13.5	4	3.17E-07	4.4584	1.10E-14	180.368	1.58625E-06	22.292	5.50E-14
13.55	4	2.46E-07	4.38525	0.0345403	180.841	1.2285E-06	21.92625	1.73E-01
13.6	2	1.90E-07	4.28948	2.58564	181.296	5.71392E-07	12.86844	7.76E+00
13.65	2	1.46E-07	4.08307	2.79765	181.735	4.38843E-07	12.24921	8.39E+00
13.7	2	1.12E-07	3.88055	3.00814	182.157	3.35766E-07	11.64165	9.02E+00
13.75	2	8.53E-08	3.68229	3.21656	182.562	2.55925E-07	11.04687	9.65E+00
13.8	2	6.48E-08	3.4888	3.42273	182.95	1.94334E-07	10.4664	1.03E+01
13.85	2	4.90E-08	3.30022	3.62599	183.321	1.47005E-07	9.90066	1.09E+01
13.9	2	3.69E-08	3.11706	3.82619	183.675	1.10786E-07	9.35118	1.15E+01
13.95	2	2.77E-08	2.93926	4.02256	184.013	8.31714E-08	8.81778	1.21E+01
14	2	2.07E-08	2.76726	4.21493	184.333	6.22068E-08	8.30178	1.26E+01
14.05	2	1.55E-08	2.60128	4.40305	184.638	4.63536E-08	7.80384	1.32E+01
14.1	2	1.15E-08	2.44143	4.58648	184.926	3.44124E-08	7.32429	1.38E+01
14.15	2	8.48E-09	2.28766	4.76447	185.199	2.54522E-08	6.86298	1.43E+01
14.2	2	6.25E-09	2.14045	4.93761	185.457	1.87581E-08	6.42135	1.48E+01
14.25	2	4.59E-09	1.99965	5.10519	185.7	1.37749E-08	5.99895	1.53E+01
14.3	2	3.36E-09	1.8654	5.26731	185.928	1.00801E-08	5.5962	1.58E+01
14.35	2	2.45E-09	1.73756	5.42348	186.143	7.35042E-09	5.21268	1.63E+01
14.4	2	1.78E-09	1.61612	5.57358	186.345	5.34144E-09	4.84836	1.67E+01
14.45	2	1.29E-09	1.50106	5.71768	186.533	3.8685E-09	4.50318	1.72E+01
14.5	2	9.31E-10	1.39219	5.8553	186.71	2.79228E-09	4.17657	1.76E+01
14.55	2	6.70E-10	1.28955	5.9872	186.874	2.009E-09	3.86865	1.80E+01
14.6	2	4.80E-10	1.19283	6.11256	187.028	1.44072E-09	3.57849	1.83E+01
14.65	2	3.43E-10	1.10191	6.23158	187.171	1.0299E-09	3.30573	1.87E+01
14.7	2	2.45E-10	1.01668	6.34476	187.304	7.33971E-10	3.05004	1.90E+01
14.75	2	1.74E-10	0.936861	6.45181	187.427	5.21463E-10	2.810583	1.94E+01
14.8	2	1.23E-10	0.862229	6.55254	187.542	3.69354E-10	2.586687	1.97E+01
14.85	2	8.70E-11	0.792691	6.64815	187.648	2.60862E-10	2.378073	1.99E+01
14.9	2	6.12E-11	0.727856	6.73746	187.746	1.83687E-10	2.183568	2.02E+01
14.95	2	4.30E-11	0.667617	6.82169	187.836	1.28979E-10	2.002851	2.05E+01

15	2	3.01E-11	0.611621	6.89975	187.92	9.0297E-11	1.834863	2.07E+01
15.05	2	2.10E-11	0.55978	6.97335	187.997	6.3045E-11	1.67934	2.09E+01
15.1	2	1.46E-11	0.511794	7.04199	188.068	4.3899E-11	1.535382	2.11E+01
15.15	2	1.02E-11	0.467408	7.10533	188.133	3.048E-11	1.402224	2.13E+01
15.2	2	7.04E-12	0.426461	7.16444	188.193	2.1105E-11	1.279383	2.15E+01
15.25	2	4.86E-12	0.388726	7.21937	188.248	1.4577E-11	1.166178	2.17E+01
15.3	2	3.35E-12	0.353995	7.27033	188.299	1.0041E-11	1.061985	2.18E+01
15.35	2	2.30E-12	0.322052	7.31729	188.345	6.897E-12	0.966156	2.20E+01
15.4	2	1.58E-12	0.292717	7.36048	188.387	4.725E-12	0.878151	2.21E+01
15.45	2	1.08E-12	0.265808	7.40025	188.426	3.231E-12	0.797424	2.22E+01
15.5	2	7.34E-13	0.241158	7.43687	188.461	2.202E-12	0.723474	2.23E+01
15.55	2	4.99E-13	0.218607	7.47082	188.494	1.497E-12	0.655821	2.24E+01
15.6	2	3.38E-13	0.197978	7.50154	188.523	1.014E-12	0.593934	2.25E+01
15.65	2	2.29E-13	0.179149	7.53001	188.55	6.87E-13	0.537447	2.26E+01
15.7	2	1.54E-13	0.161965	7.55581	188.574	4.62E-13	0.485895	2.27E+01
15.75	2	1.04E-13	0.146306	7.57955	188.597	3.12E-13	0.438918	2.27E+01
15.8	2	7.00E-14	0.132047	7.60114	188.617	2.1E-13	0.396141	2.28E+01
15.85	2	4.70E-14	0.119076	7.62081	188.635	1.41E-13	0.357228	2.29E+01
15.9	2	3.10E-14	0.107289	7.63873	188.652	9.3E-14	0.321867	2.29E+01
15.95	2	2.10E-14	0.0965867	7.65502	188.667	6.3E-14	0.2897601	2.30E+01
16	2	1.40E-14	0.0868799	7.66999	188.681	4.2E-14	0.2606397	2.30E+01
16.05	2	9.00E-15	0.0780818	7.6835	188.693	2.7E-14	0.2342454	2.31E+01
16.1	2	6.00E-15	0.0701136	7.69558	188.704	1.8E-14	0.2103408	2.31E+01
16.15	2	4.00E-15	0.0629064	7.70666	188.715	1.2E-14	0.1887192	2.31E+01

time	Temp. after compress °C	Vap frac. after compr °C	Temp. after first heat e °C	Vap frac. Af °C	Heat Duty of seco [MJ/hr]	Heat D. Intercooler [MJ/hr]
0	193.6109009	1	-270.3722229	0	999.9386597	0.002782357
0.05	177.2104797	1	115.1567459	0	-169.3969574	24.07682419
0.1	177.2103424	1	115.072319	0	-168.855545	24.06489372
0.15	177.2101135	1	114.9708939	0	-168.2058411	24.05089188
0.2	177.2098694	1	114.8682556	0	-167.5495605	24.03661919
0.25	177.2096252	1	114.7782288	0	-166.9747772	24.0241642
0.3	177.2094269	1	114.6889114	0	-166.4049072	24.01169205
0.35	177.2092285	1	114.6061478	0	-165.8776398	24.00061417
0.4	177.2089844	1	114.5205612	0	-165.3331757	23.98855209
0.45	177.208786	1	114.4442673	0	-164.8483124	23.97802925
0.5	177.2086487	1	114.3593216	0	-164.3091583	23.96640968
0.55	177.2084503	1	114.2821121	0	-163.8194885	23.95578384
0.6	177.2082367	1	114.2060242	0	-163.3374634	23.94542503
0.65	177.2080383	1	114.1267471	0	-162.8359528	23.93466187
0.7	177.207901	1	114.0583878	0	-162.4036865	23.92526054
0.75	177.2077637	1	113.975853	0	-161.8827515	23.91400146
0.8	177.2076263	1	113.8983765	0	-161.3939819	23.90351295
0.85	177.207428	1	113.8219757	0	-160.9126892	23.89308167
0.9	177.2072601	1	113.7695236	0	-160.5824432	23.88596153
0.95	177.2071228	1	113.6890945	0	-160.0766602	23.8751297
1	177.2069855	1	113.6168976	0	-159.6230469	23.86525726
1.05	177.2068481	1	113.5351791	0	-159.1102753	23.85438156
1.1	177.2067108	1	113.4730911	0	-158.7211761	23.84594727
1.15	177.206543	1	113.4054489	0	-158.2975159	23.83691788
1.2	177.2064514	1	113.3433304	0	-157.9085846	23.82859421
1.25	177.2063141	1	113.262558	0	-157.403656	23.81768227
1.3	177.2061462	1	113.200676	0	-157.0173187	23.80940628
1.35	177.2060699	1	113.1368256	0	-156.6190948	23.80078697

1.4	177.2059326	1	113.0594788	0	-156.1367645	23.79053116
1.45	177.2058411	1	112.9911194	0	-155.7112122	23.78158379
1.5	177.205658	1	112.9251328	0	-155.3005066	23.77280426
1.55	177.2055969	1	112.8567429	0	-154.8755188	23.76363754
1.6	177.2054596	1	112.7838364	0	-154.4230042	23.7541256
1.65	177.2053986	1	112.700592	0	-153.9065247	23.74316978
1.7	177.2052917	1	112.6408157	0	-153.536087	23.73522186
1.75	177.2051544	1	112.5633621	0	-153.056778	23.7250576
1.8	177.2050476	1	112.4990768	0	-152.6592102	23.71665382
1.85	177.2049561	1	112.4155273	0	-152.1429749	23.70571709
1.9	177.2048798	1	112.3403168	0	-151.6789856	23.69586372
1.95	177.2047882	1	112.2726669	0	-151.2617798	23.68695259
2	177.2046814	1	112.1953201	0	-150.7855377	23.67694092
2.05	177.2045746	1	112.115097	0	-150.2917938	23.66657066
2.1	177.2045135	1	112.0283585	0	-149.7589874	23.65520477
2.15	177.2044373	1	111.9596252	0	-149.337204	23.64630318
2.2	177.2043457	1	111.8815994	0	-148.8585815	23.63625526
2.25	177.2042389	1	111.8026657	0	-148.3752594	23.62600136
2.3	177.2041321	1	111.7155838	0	-147.8425598	23.61483192
2.35	177.2040405	1	111.6324768	0	-147.3345032	23.60415649
2.4	177.2039337	1	111.5589676	0	-146.8859406	23.59471703
2.45	177.2039032	1	111.466362	0	-146.3213806	23.58279037
2.5	177.2038269	1	111.3939667	0	-145.8805389	23.57350731
2.55	177.2037659	1	111.2861404	0	-145.224472	23.55980492
2.6	177.2036591	1	111.208252	0	-144.7514954	23.54985809
2.65	177.2036285	1	111.1136093	0	-144.1773987	23.5378418
2.7	177.2035217	1	111.024971	0	-143.6401978	23.52659607
2.75	177.2034302	1	110.9353867	0	-143.0982666	23.51522636
2.8	177.2033997	1	110.8412552	0	-142.5293121	23.50331497
2.85	177.2032928	1	110.7514725	0	-141.9873962	23.49194717
2.9	177.2032623	1	110.6497116	0	-141.3740234	23.47912979
2.95	177.2031555	1	110.549881	0	-140.7728424	23.46656799
3	177.2030487	1	110.4511414	0	-140.179245	23.45413399
3.05	177.2030182	1	110.35186	0	-139.5834198	23.44166565

3.1	177.2029572	1	110.2343292	0	-138.8789215	23.42708206
3.15	177.2029114	1	110.1417923	0	-138.3250885	23.41546822
3.2	177.2028198	1	110.0239334	0	-137.6206055	23.40076256
3.25	177.2027893	1	109.9220657	0	-137.0127258	23.38806534
3.3	177.2026825	1	109.8087845	0	-136.3378754	23.37397385
3.35	177.202652	1	109.6970901	0	-135.6732635	23.36008835
3.4	177.2026062	1	109.5827103	0	-134.9938812	23.34601402
3.45	177.2025146	1	109.4797668	0	-134.3834076	23.33314705
3.5	177.2024841	1	109.3559036	0	-133.6498108	23.31795311
3.55	177.2023773	1	109.233017	0	-132.9234467	23.30279922
3.6	177.2023468	1	109.107048	0	-132.1800995	23.28739929
3.65	177.2022705	1	108.986702	0	-131.4711609	23.27260971
3.7	177.202179	1	108.8590393	0	-130.7204895	23.25707054
3.75	177.2021332	1	108.7307281	0	-129.9672852	23.24137688
3.8	177.2021027	1	108.6026611	0	-129.2167053	23.22582436
3.85	177.2020416	1	108.4646149	0	-128.4092712	23.20892334
3.9	177.2019958	1	108.3288498	0	-127.6167221	23.19251251
3.95	177.2019043	1	108.1852112	0	-126.7797546	23.17508507
4	177.2018738	1	108.0464249	0	-125.9724808	23.15839005
4.05	177.201828	1	107.9056702	0	-125.1554794	23.14138985
4.1	177.2017975	1	107.7556915	0	-124.2866898	23.123312
4.15	177.2017365	1	107.6134796	0	-123.4646149	23.10630608
4.2	177.2016296	1	107.4611969	0	-122.5858994	23.08810616
4.25	177.2015991	1	107.2931519	0	-121.6185455	23.06801987
4.3	177.2015686	1	107.1367645	0	-120.7203827	23.04933167
4.35	177.2014923	1	106.9722748	0	-119.7776718	23.02984619
4.4	177.2014618	1	106.8147659	0	-118.8770676	23.01121521
4.45	177.201355	1	106.6348495	0	-117.8504868	22.98989296
4.5	177.2013245	1	106.4653397	0	-116.8857727	22.9698658
4.55	177.2012939	1	106.2876663	0	-115.8767624	22.94889069
4.6	177.2012634	1	106.1243286	0	-114.9515152	22.92977524
4.65	177.2011566	1	105.9313507	0	-113.8610001	22.90725517
4.7	177.2011261	1	105.7497406	0	-112.8374557	22.88607597
4.75	177.2011261	1	105.5645981	0	-111.7965012	22.86438751

4.8	177.2010193	1	105.3754272	0	-110.7356491	22.84238243
4.85	177.2009583	1	105.1646194	0	-109.5565796	22.8180027
4.9	177.2009125	1	104.9669266	0	-108.4540482	22.79523659
4.95	177.200882	1	104.7580566	0	-107.2923737	22.77114296
5	177.2008209	1	104.5485687	0	-106.1306381	22.74707031
5.05	177.2007751	1	104.3251419	0	-104.8950577	22.72144318
5.1	177.2007446	1	104.1157913	0	-103.7408676	22.69760704
5.15	177.2006836	1	103.89505	0	-102.5275192	22.67255783
5.2	177.2006073	1	103.6677933	0	-101.2821198	22.6467514
5.25	177.2005768	1	103.4163666	0	-99.90856934	22.6182785
5.3	177.2005157	1	103.1781235	0	-98.61140442	22.59151077
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6.05	177.1996918	1	98.82709503	0	-75.63787079	22.11663055
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6.2	177.1995544	1	97.73962402	0	-70.10124969	22.00222397
6.25	177.1995239	1	97.36170959	0	-68.19589996	21.96275711
6.3	177.1994629	1	96.98044586	0	-66.28316498	21.92334557
6.35	177.1994324	1	96.59728241	0	-64.37055206	21.88380051
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6.95	177.1987152	1	90.87323761	0	-36.90997696	21.3169384
7	177.1986389	1	90.30384064	0	-34.28788757	21.26276207
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7.1	177.1985474	1	89.1246109	0	-28.91847038	21.15200043
7.15	177.1985168	1	88.50273132	0	-26.11924744	21.09426498
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7.3	177.1983337	1	86.54346466	0	-17.44334602	20.91526222
7.35	177.1982727	1	85.84400177	0	-14.39780712	20.85240555
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7.8	177.1977997	1	78.48960876	0	16.06019974	20.22448349
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7.9	177.1976624	1	76.53590393	0	23.69673347	20.06716728
7.95	177.1976929	1	75.5220108	0	27.5885849	19.98692322
8	177.1976624	1	74.4634552	0	31.60115242	19.90426826
8.05	177.1976318	1	73.39181519	0	35.61038971	19.82177925
8.1	177.1975555	1	72.24909973	0	39.82950974	19.73487854
8.15	177.1974945	1	71.10147095	0	44.00894165	19.6488018

8.2	177.1975555	1	69.86954498	0	48.43193817	19.55762672
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8.35	177.197525	1	65.997612	0	61.91934586	19.28009987
8.4	177.197464	1	64.62537384	0	66.55371857	19.18477058
8.45	177.197525	1	63.15537262	0	71.43585205	19.08433914
8.5	177.1976013	1	61.72230148	0	76.11590576	18.98809433
8.55	177.1975555	1	60.18536758	0	81.04943848	18.8866787
8.6	177.1976318	1	58.58387375	0	86.09830475	18.78291512
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8.7	177.1979065	1	55.25237274	0	96.3081131	18.57324982
8.75	177.1980286	1	53.46224213	0	101.6366425	18.46406746
8.8	177.1983337	1	51.63271332	0	106.9722366	18.35482788
8.85	177.1988525	1	49.74225235	0	112.3713684	18.24467278
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9	180.311615	1	43.45315552	0	120.1175995	23.96788788
9.05	183.4264069	1	41.81497192	0	117.6058884	28.88408279
9.1	193.6128693	1	98.71665955	0	-66.57486725	48.38911819
9.15	193.6130066	1	98.05029297	0	-63.94444275	48.26996231
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9.25	196.459198	1	98.24255371	0	-62.69842148	50.88578033
9.3	203.0704956	1	101.0635986	0	-68.8225708	56.15716934
9.35	211.4131927	1	104.1904602	0	-75.01062775	61.44734573
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9.6	257.5636292	1	122.7634964	0	-105.7596664	83.1533432
9.65	266.5106506	1	126.5382767	0	-110.828743	86.8024292
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9.9	306.778656	1	143.9635773	0	-129.2310486	101.1688919
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11.95	367.335022	1	167.2637787	0	-119.2246475	116.8472595
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12.3	367.3719788	1	164.3310394	0	-113.2864914	116.2384033
12.35	367.3742371	1	163.9062042	0	-112.4374466	116.1500702
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12.65	367.3813477	1	160.9600677	0	-106.6046219	115.5430908
12.7	367.381897	1	160.4467621	0	-105.5967712	115.4384079
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12.8	367.3826294	1	159.350708	0	-103.4525528	115.2144699
12.85	367.382843	1	158.8427582	0	-102.4623032	115.1109772
12.9	367.3831177	1	158.2460022	0	-101.3017731	114.9905472
12.95	367.3831787	1	157.6993103	0	-100.2411118	114.8793335
13	367.3835144	1	157.0947571	0	-99.07102203	114.757576
13.05	367.3835754	1	156.5151825	0	-97.95220184	114.6404343
13.1	367.3836365	1	155.8885651	0	-96.74549103	114.5150223
13.15	367.383728	1	155.2489777	0	-95.5171051	114.3864975
13.2	367.3837891	1	154.6154022	0	-94.30329895	114.2600708
13.25	367.383728	1	153.9634705	0	-93.05773163	114.1293793

13.3	367.3837891	1	153.3008728	0	-91.79528046	113.998085
13.35	367.3838501	1	152.630127	0	-90.52064514	113.8643494
13.4	367.3838501	1	151.9420624	0	-89.21673584	113.7286606
13.45	367.3839111	1	151.2327576	0	-87.87639618	113.588974
13.5	367.3839111	1	150.5200806	0	-86.53353119	113.4485855
13.55	373.5588379	1	151.2781982	0	-87.54368591	116.9095154
13.6	321.4978333	1	181.0343933	0	-143.6678314	76.22077179
13.65	318.5427856	1	184.6828156	0	-150.9433441	73.01770782
13.7	315.8691711	1	188.2410278	0	-158.1621552	69.86045074
13.75	313.4510193	1	191.6974182	0	-165.288269	66.75937653
13.8	311.2644043	1	195.072113	0	-172.3768005	63.72338104
13.85	309.2869263	1	198.3374329	0	-179.3345795	60.75603867
13.9	307.4988708	1	201.5149841	0	-186.2223663	57.86685944
13.95	305.8817139	1	204.5668488	0	-192.8991547	55.05499268
14	304.4192505	1	207.5118561	0	-199.414505	52.32880402
14.05	303.0961914	1	210.3540955	0	-205.7702942	49.69319916
14.1	301.8995667	1	213.0847778	0	-211.9204865	47.14942551
14.15	300.8170166	1	215.6760559	0	-217.7398682	44.69771194
14.2	299.8372498	1	218.1826019	0	-223.4305573	42.34759521
14.25	298.9505615	1	220.5750732	0	-228.864624	40.09652328
14.3	298.1478882	1	222.8711853	0	-234.1068726	37.94799423
14.35	297.4211426	1	225.0540466	0	-239.0749512	35.89956665
14.4	296.7627869	1	227.1270294	0	-243.7740021	33.95241547
14.45	296.1672974	1	229.1012115	0	-248.2456818	32.10494232
14.5	295.6275024	1	230.9593964	0	-252.4030151	30.35580444
14.55	295.1384583	1	232.739212	0	-256.417511	28.70628929
14.6	294.6954041	1	234.4078522	0	-260.1264038	27.15096474
14.65	294.6546326	0.998775065	235.9760132	0	-263.5743103	25.68877792
14.7	295.1001282	0.995985746	237.4648895	0	-266.861908	24.31720352
14.75	295.5115662	0.993374705	238.8616486	0	-269.9186401	23.03254128
14.8	295.8909302	0.990947008	240.159668	0	-272.7023621	21.83037186
14.85	296.2406006	0.988680661	241.40065	0	-275.4276123	20.71118164
14.9	296.5623779	0.986581147	242.5400391	0	-277.8482971	19.6668663
14.95	296.8583374	0.984633803	243.6184998	0	-280.1765137	18.69710159

15	297.1302185	0.982828438	244.5962372	0	-282.1832886	17.79479408
15.05	297.3797913	0.981158614	245.5276337	0	-284.1619873	16.95993233
15.1	297.6086731	0.979616225	246.3922424	0	-285.9890137	16.19366646
15.15	297.8183594	0.978194594	247.1762695	0	-287.5744324	15.47935581
15.2	298.0103455	0.976881504	247.9102173	0	-289.0818481	14.82279015
15.25	298.1860657	0.975682855	248.5918427	0	-290.4883118	14.21297264
15.3	298.3465881	0.974577248	249.2237854	0	-291.7988281	13.65301704
15.35	298.4933472	0.973564327	249.8016663	0	-292.9758301	13.13643837
15.4	298.6271973	0.972637653	250.3291321	0	-294.029541	12.66506577
15.45	298.7493591	0.971782625	250.8130493	0	-294.9897766	12.22649574
15.5	298.8606873	0.971006513	251.2580566	0	-295.8735046	11.82717037
15.55	298.9620972	0.97029382	251.6747284	0	-296.7344971	11.46115208
15.6	299.0543823	0.969651759	252.0439606	0	-297.4462891	11.12675095
15.65	299.1383362	0.96905607	252.3907013	0	-298.1516418	10.82505894
15.7	299.2146301	0.968520463	252.7003937	0	-298.7512817	10.54525661
15.75	299.2839355	0.968025506	252.9873962	0	-299.324646	10.2831049
15.8	299.3468933	0.967582524	253.2467499	0	-299.8314209	10.05102253
15.85	299.4039917	0.967176199	253.4825134	0	-300.2910767	9.839715958
15.9	299.455719	0.966808736	253.6970978	0	-300.7095947	9.647149086
15.95	299.502594	0.966475129	253.8917999	0	-301.0874939	9.467474937
16	299.5449829	0.966179609	254.0730286	0	-301.4584045	9.313630104
16.05	299.5833435	0.96589905	254.2352142	0	-301.7808533	9.17149353
16.1	299.6180725	0.965651453	254.3776398	0	-302.0441284	9.039224625
16.15	299.6180725	0.965651453	254.3776398	0	-302.0441284	9.039224625

[MJ/hr]	[MJ/hr]	°C	°C	°C	°C	
Heat D. Comp.1	Heat D. Comp.2	Temp. of inter	LMTD	T1 after first cor	T2 after interco	F safety factor
0.005245263	0.005519986	25.0029068	107.0683823	145.5064392	119.6709747	1
97.6080246	99.72086334	50.05176926	89.29512024	131.6281433	122.4860535	1
97.56270599	99.67485809	50.03936005	89.30162811	131.6278992	122.4861526	1
97.50853729	99.61934662	50.02477646	89.30931854	131.6277008	122.4862595	1
97.45367432	99.56330109	50.00992584	89.31712341	131.6274567	122.4863586	1
97.40570831	99.51416016	49.99697113	89.32395935	131.6272888	122.4864578	1
97.35806274	99.46547699	49.9839859	89.3307724	131.6270447	122.4865646	1
97.31420898	99.4204483	49.97245789	89.33691406	131.6270142	122.4866333	1
97.26863098	99.37387085	49.95987701	89.34349823	131.6267395	122.4867325	1
97.22808838	99.33244324	49.94892502	89.34925842	131.6265717	122.4868011	1
97.18312073	99.2864151	49.9368515	89.35561371	131.6264038	122.4869003	1
97.1421814	99.24459076	49.92580032	89.36142731	131.626236	122.486969	1
97.10195923	99.20339203	49.91501617	89.36711121	131.6260986	122.4870377	1
97.06009674	99.16052246	49.90379333	89.37302399	131.6259613	122.4871063	1
97.02400208	99.12365723	49.89402771	89.37817383	131.625824	122.487175	1
96.98049927	99.07913208	49.8822937	89.38437653	131.6257019	122.4872742	1
96.93968201	99.0374527	49.87137604	89.39012146	131.6255646	122.4873428	1
96.8994751	98.99624634	49.86052704	89.39582825	131.6253967	122.4874115	1
96.87192535	98.96804047	49.85309982	89.39971161	131.6252594	122.487442	1
96.8296814	98.92488861	49.84184265	89.40566254	131.6251526	122.4875107	1
96.79174805	98.88613892	49.83156586	89.41105652	131.6249847	122.4875793	1
96.74897766	98.84237671	49.82024384	89.41704559	131.6248779	122.487648	1
96.71648407	98.80912781	49.81145859	89.42168427	131.6247864	122.4877167	1
96.68113708	98.77293396	49.8020668	89.42660522	131.624649	122.4877472	1
96.64868164	98.73970795	49.79338837	89.43121338	131.6245728	122.4878159	1
96.60645294	98.69660187	49.7820282	89.43717957	131.6244049	122.4878845	1
96.57418823	98.66355896	49.77341461	89.4417038	131.6242981	122.4879227	1
96.54090118	98.62960052	49.76446152	89.44641876	131.624176	122.4879837	1

96.50062561	98.58839417	49.75378036	89.45207977	131.6240997	122.4880524	1
96.46517944	98.55210876	49.74445724	89.45700836	131.6240387	122.4880905	1
96.43087769	98.51698303	49.7353363	89.46179199	131.6239014	122.488121	1
96.39533997	98.48073578	49.72577286	89.46683502	131.6237946	122.4881897	1
96.35764313	98.44203186	49.71587372	89.47206116	131.6237335	122.4882278	1
96.3144989	98.39801025	49.70447922	89.47810364	131.6236572	122.4882889	1
96.2835083	98.36639404	49.69620514	89.48243713	131.6235199	122.488327	1
96.24354553	98.32540894	49.68562698	89.48804474	131.6234589	122.4883957	1
96.21037292	98.29147339	49.67687607	89.49266052	131.6233826	122.4884262	1
96.16724396	98.24743652	49.66548538	89.49866486	131.623291	122.4884644	1
96.12851715	98.20782471	49.65524292	89.50408936	131.6232147	122.488533	1
96.09362793	98.17222595	49.64595413	89.5089798	131.6231232	122.4885635	1
96.05389404	98.13157654	49.63554382	89.51447296	131.6230469	122.4885941	1
96.01270294	98.08945465	49.62475967	89.52017212	131.6229858	122.4886322	1
95.96815491	98.04399872	49.61292648	89.52642822	131.622879	122.4887009	1
95.93293762	98.00798035	49.60363388	89.53131104	131.6227722	122.4887314	1
95.893013	97.96711731	49.59319305	89.53684235	131.6227417	122.4887695	1
95.8526535	97.9258728	49.5825119	89.54247284	131.6226501	122.4888	1
95.80820465	97.88037872	49.57088089	89.5486145	131.6225739	122.4888382	1
95.76579285	97.83700562	49.55979156	89.55446625	131.6225128	122.4888687	1
95.72834015	97.79869843	49.54995728	89.55966187	131.6224365	122.4888992	1
95.68119812	97.75051117	49.53754807	89.5662384	131.6223755	122.4889679	1
95.64437103	97.71294403	49.52788544	89.57133484	131.6222992	122.489006	1
95.58963013	97.65698242	49.51360703	89.57888031	131.6222382	122.4890366	1
95.55012512	97.61657715	49.50326538	89.58434296	131.6221619	122.4890747	1
95.50218964	97.5676651	49.49075317	89.59095764	131.6221008	122.4891052	1
95.45741272	97.52177429	49.47905731	89.59713745	131.6220398	122.4891434	1
95.41214752	97.47548676	49.46722031	89.6033783	131.6219635	122.4891739	1
95.36463165	97.42700195	49.4548111	89.60995483	131.621933	122.4892044	1
95.31937408	97.38071442	49.44297791	89.61619568	131.6218567	122.4892426	1
95.26815033	97.32844543	49.42964935	89.62324524	131.6217957	122.4892731	1
95.21803284	97.27713013	49.41656113	89.63014984	131.6217346	122.4893112	1
95.16845703	97.22642517	49.40361023	89.63699341	131.6216583	122.4893417	1
95.11869049	97.17564392	49.39065552	89.64383698	131.6215973	122.4893799	1

95.05995941	97.11551666	49.37546539	89.65189362	131.6215973	122.4894104	1
95.01367188	97.0683136	49.36336136	89.65828705	131.621521	122.4894485	1
94.95484924	97.00817108	49.34806824	89.66636658	131.62146	122.4894791	1
94.90413666	96.9563446	49.33484268	89.6733551	131.6213837	122.4895096	1
94.84778595	96.89870453	49.32019424	89.68109894	131.6213531	122.4895477	1
94.79225922	96.84204865	49.30571747	89.68874359	131.6212921	122.4895782	1
94.73560333	96.78413391	49.29106903	89.69650269	131.6212463	122.4896164	1
94.68457794	96.731987	49.27767181	89.70355225	131.6211548	122.4896469	1
94.62339783	96.66944885	49.26187134	89.71192932	131.6211548	122.4896851	1
94.56272888	96.60741425	49.24610519	89.72026062	131.6210785	122.4897156	1
94.50071716	96.54405212	49.23006439	89.72875214	131.621048	122.4897537	1
94.4414978	96.48361206	49.21467209	89.73684692	131.6209412	122.4897537	1
94.37887573	96.419487	49.19849777	89.74541473	131.6209412	122.4897842	1
94.31595612	96.35527802	49.18215179	89.75404358	131.6208801	122.4898148	1
94.25334167	96.2912674	49.16597748	89.76262665	131.6208801	122.4898529	1
94.18589783	96.22234344	49.14838028	89.77189636	131.6207733	122.4898834	1
94.11977386	96.15477753	49.13129044	89.78093719	131.6207428	122.4899216	1
94.04987335	96.08332825	49.11315155	89.79051208	131.6206818	122.4899521	1
93.98252869	96.01447296	49.09578705	89.79971313	131.6206818	122.4899902	1
93.91435242	95.94481659	49.07808685	89.80905151	131.6206055	122.4900208	1
93.84178162	95.87073517	49.05926895	89.81898499	131.6205444	122.4900589	1
93.77320862	95.80056	49.04156876	89.82836151	131.6205444	122.4900894	1
93.69981384	95.7256012	49.02261353	89.83834839	131.6204681	122.4900894	1
93.61910248	95.64312744	49.00169373	89.84939575	131.6204071	122.4901199	1
93.54405975	95.56653595	48.98226166	89.85967255	131.6203766	122.4901581	1
93.46541595	95.48606873	48.96198654	89.87038422	131.6203308	122.4901886	1
93.39026642	95.40927887	48.94258881	89.88064575	131.6203003	122.4902267	1
93.30454254	95.32164001	48.92037964	89.89237213	131.6202698	122.4902573	1
93.22403717	95.23937225	48.8995285	89.90338135	131.6202087	122.4902954	1
93.13975525	95.15335083	48.87768936	89.91490173	131.6201324	122.4903259	1
93.06255341	95.07445526	48.8578186	89.92541504	131.6201324	122.4903641	1
92.97158051	94.98139191	48.8343544	89.93781281	131.6201019	122.4903946	1
92.88614655	94.89408875	48.81231308	89.94946289	131.6200714	122.4904251	1
92.79920197	94.80535126	48.78973007	89.96137238	131.6199951	122.4904633	1

92.71061707	94.71479034	48.76684189	89.97344208	131.6199341	122.4904938	1
92.61222839	94.61423492	48.74144745	89.98686218	131.6199036	122.4905319	1
92.52023315	94.52022552	48.71774292	89.99938965	131.6199036	122.4905624	1
92.42321777	94.42118835	48.69268417	90.01260376	131.6198273	122.4906006	1
92.32627869	94.32204437	48.66762543	90.02584076	131.6197968	122.4906311	1
92.22309113	94.21669769	48.64094162	90.03990936	131.6197205	122.4906693	1
92.12677765	94.1182785	48.61611938	90.05300903	131.6196899	122.4906998	1
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91.92153168	93.9085083	48.56319046	90.08094788	131.6196289	122.4907684	1
91.80683136	93.79139709	48.53355408	90.09657288	131.6195526	122.490799	1
91.69860077	93.6807251	48.50568008	90.11126709	131.6195221	122.4908371	1
91.58708954	93.5667572	48.47699356	90.1264267	131.6195221	122.4908676	1
91.47424316	93.45153809	48.44786835	90.14176941	131.6194611	122.4909058	1
91.35778809	93.33242798	48.41778946	90.15763855	131.6194153	122.4909363	1
91.23795319	93.21009064	48.38676453	90.1740036	131.6193542	122.4910049	1
91.11923981	93.08880615	48.35621262	90.19011688	131.6193237	122.4910355	1
91.00312805	92.97012329	48.32623672	90.2059021	131.6192474	122.4910736	1
90.87310028	92.83727264	48.29277039	90.22355652	131.6192474	122.4911041	1
90.74998474	92.71147156	48.26099777	90.24029541	131.6191864	122.4911423	1
90.61812592	92.57672119	48.2270546	90.25817871	131.6191559	122.4911728	1
90.48493958	92.44071198	48.19270706	90.27626801	131.6190796	122.4912109	1
90.34579468	92.29853821	48.15673065	90.29524231	131.6190491	122.4912796	1
90.21166992	92.16149139	48.12224579	90.3134079	131.6190186	122.4913101	1
90.06554413	92.01215363	48.0846405	90.33322144	131.618988	122.4913406	1
89.92443085	91.86797333	48.04835892	90.3523407	131.618988	122.4913788	1
89.78098297	91.72136688	48.01136398	90.37180328	131.6189117	122.4914093	1
89.62766266	91.56481171	47.9717598	90.39266205	131.6188507	122.491478	1
89.47400665	91.40771484	47.93225479	90.41345215	131.6188049	122.4915161	1
89.31884766	91.24918365	47.89227676	90.43449402	131.6187744	122.4915466	1
89.15976715	91.08670807	47.85118103	90.45613098	131.6187134	122.4916153	1
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88.84047699	90.76047516	47.76898575	90.49938965	131.6186371	122.4917145	1
88.66703033	90.58322906	47.72432709	90.52285767	131.618576	122.4917526	1
88.49500275	90.40755463	47.68001175	90.54614258	131.6184998	122.4917831	1

88.31855011	90.22718048	47.63457489	90.57007599	131.6184998	122.4918518	1
88.13544464	90.04005432	47.58750916	90.59481812	131.6184692	122.49189	1
87.95520782	89.85601807	47.54101944	90.61925507	131.6184082	122.491951	1
87.76386261	89.66049194	47.49175262	90.64512634	131.6183319	122.4919891	1
87.57624817	89.46876526	47.44343185	90.67053223	131.6183014	122.4920578	1
87.38018799	89.26844025	47.39297485	90.69702148	131.6182404	122.4920883	1
87.17186737	89.05561066	47.33936691	90.72520447	131.6182404	122.492157	1
86.97438812	88.85381317	47.28853607	90.75190735	131.6181946	122.4922256	1
86.7596283	88.63438416	47.23320007	90.78096008	131.6181335	122.4922943	1
86.54837036	88.41850281	47.17891312	90.8094635	131.618103	122.4923248	1
86.32949829	88.19487762	47.12252045	90.83905029	131.6180267	122.4923935	1
86.10756683	87.96821594	47.06542206	90.86901855	131.6179962	122.4924622	1
85.88130951	87.7370224	47.00719833	90.89953613	131.6179352	122.4925003	1
85.64765167	87.49828339	46.94711304	90.93106842	131.6178894	122.4925613	1
85.41139984	87.25682831	46.88641739	90.96292114	131.6178894	122.49263	1
85.16699219	87.00717926	46.82338333	90.99594116	131.6177979	122.4926987	1
84.92347717	86.75836945	46.76075363	91.02877045	131.6177673	122.4927673	1
84.66928101	86.4986496	46.69534302	91.06305695	131.6177216	122.492836	1
84.40789032	86.23159027	46.62810516	91.09828186	131.617691	122.4929047	1
84.1463089	85.96430206	46.56086349	91.13350677	131.6176605	122.4929733	1
83.87320709	85.68527985	46.49067307	91.1702652	131.61763	122.493042	1
83.5983963	85.4046402	46.4200058	91.20722961	131.6175537	122.4930725	1
83.31769562	85.11787415	46.34767914	91.24508667	131.6174927	122.4931717	1
83.02405548	84.81785583	46.27223587	91.28455353	131.6174622	122.4932404	1
82.72570038	84.51302338	46.19556808	91.32466125	131.6174164	122.493309	1
82.42610931	84.20692444	46.11856079	91.36492157	131.6173859	122.4933777	1
82.12692261	83.90124512	46.04165649	91.40512085	131.6173553	122.4934464	1
81.80268097	83.56996155	45.95837784	91.44864655	131.6173248	122.493515	1
81.4895401	83.24997711	45.87791443	91.49067688	131.6172791	122.4935837	1
81.16468811	82.91814423	45.79436493	91.53432465	131.6172485	122.4936829	1
80.82976532	82.57604218	45.70830536	91.57923889	131.617218	122.4937515	1
80.49514008	82.23406982	45.62244797	91.62408447	131.6172485	122.4938202	1
80.14296722	81.87425995	45.53197861	91.67127991	131.617218	122.4938889	1
79.7940979	81.51780701	45.44235992	91.71800995	131.6171875	122.4939575	1

79.42486572	81.14070892	45.34745026	91.76750183	131.617157	122.4940567	1
79.05780029	80.76559448	45.25328827	91.81661224	131.6171875	122.4941254	1
78.68024445	80.37985229	45.15641022	91.86708832	131.6171875	122.494194	1
78.2991333	79.99059296	45.05855179	91.91805267	131.6171875	122.4942627	1
77.91234589	79.5953598	44.95930099	91.96973419	131.6171875	122.4943314	1
77.50482178	79.179039	44.85475922	92.02418518	131.6172485	122.4944305	1
77.11422729	78.78000641	44.75456238	92.07633972	131.6172791	122.4944992	1
76.70240784	78.35927582	44.64900589	92.13121033	131.6172791	122.4945297	1
76.28101349	77.92878723	44.54097366	92.18740082	131.6173553	122.4945984	1
75.85334778	77.49184418	44.43144989	92.24436951	131.6174927	122.4946671	1
75.42879486	77.0582428	44.32270432	92.30088806	131.6175842	122.4946976	1
74.98407745	76.60388184	44.2090416	92.3599472	131.6177673	122.4946976	1
74.53874969	76.14897156	44.09531403	92.41905212	131.6179962	122.4946976	1
74.08810425	75.68878174	43.9806366	92.47865295	131.6184082	122.4945984	1
73.63122559	75.22225189	43.86585617	92.53849792	131.6194153	122.4942627	1
73.31827545	75.00643158	44.75310516	92.19654846	132.0110931	122.3180008	1
73.74188232	75.97717285	49.93837738	90.25382996	134.2877045	121.4276352	1
74.31585693	77.08005524	55.05475235	88.58869171	136.9406433	120.647644	1
91.21907043	95.99689484	75.32249451	81.818573	145.5080719	119.6710129	1
90.994133	95.76013184	75.19889832	81.88710022	145.5082397	119.6710129	1
90.75814819	95.51186371	75.0695343	81.95890808	145.5086517	119.6710129	1
91.22792053	96.28478241	77.91139984	81.74156952	147.8746796	119.7729416	1
92.47431183	98.14853668	83.37252045	82.13748169	153.3570251	120.6220093	1
93.89387512	100.1341705	88.84554291	84.11064148	160.589447	122.8228683	1
95.38558197	102.2046051	93.80420685	86.88551331	168.3753967	125.4861679	1
96.85870361	104.2056198	98.4903183	89.72654724	176.1836243	127.9423904	1
98.30282593	106.1175995	99.96736908	94.15620422	183.9524841	130.1170807	1
99.68465424	107.9105072	99.96736908	99.15026855	191.6452332	132.0180817	1
101.01548	109.6039047	99.96736908	103.8932724	199.2257233	133.6702728	1
102.294632	111.2048798	99.96736908	108.3832855	206.649765	135.1021576	1
103.4982071	112.6926956	99.96736908	112.6184998	213.8742981	136.3398132	1
104.6368484	114.0843811	99.96736908	116.5927963	220.8477325	137.4078217	1
105.709137	115.3831635	99.96736908	120.300766	227.5213928	138.3278351	1
106.7275848	116.6052704	99.96736908	123.7394638	233.852829	139.1194611	1

107.6659851	117.7247314	99.96736908	126.9062653	239.8033142	139.7992554	1
108.5565262	118.7797623	99.96736908	129.8028107	245.3449097	140.3822784	1
109.3843536	119.7555313	99.96736908	132.4337616	250.4589691	140.8816833	1
110.155426	120.6647415	99.96736908	134.806778	255.1366119	141.3087463	1
110.8586502	121.4877319	99.96736908	136.9320679	259.377594	141.6733398	1
111.5124741	122.2498779	99.96736908	138.8234253	263.1920471	141.9844055	1
112.1046829	122.9382858	99.96736908	140.4962006	266.5973816	142.2491302	1
112.6461334	123.5657196	99.96736908	141.9675903	269.6164856	142.4746552	1
113.1429749	124.1373291	99.96736908	143.2547607	272.2760925	142.6661682	1
113.5828476	124.6471558	99.96736908	144.3751373	274.6052551	142.8283997	1
113.9735413	125.0945969	99.96736908	145.3468475	276.6350403	142.966568	1
114.3216705	125.4975967	99.96736908	146.1860809	278.3960571	143.0834503	1
114.6156921	125.836319	99.96736908	146.9079285	279.9164429	143.1823273	1
114.8796158	126.1392365	99.96736908	147.5275269	281.225647	143.2660217	1
115.1035767	126.3965149	99.96736908	148.0575867	282.3486938	143.3367462	1
115.2926788	126.6134949	99.96736908	148.5101929	283.309967	143.3964539	1
115.451416	126.7961655	99.96736908	148.8955841	284.1299744	143.4469147	1
115.5852051	126.9501114	99.96736908	149.223114	284.828064	143.4894409	1
115.6961899	127.0775757	99.96736908	149.5012817	285.4218445	143.5253143	1
114.9060898	126.2189026	99.96736908	149.9363098	286.3515015	143.5813293	1
114.9420853	126.2621002	99.96736908	150.1027527	286.7078552	143.6025238	1
114.9549789	126.2794495	99.96736908	150.2435608	287.0093994	143.6204529	1
114.9712143	126.2995911	99.96736908	150.3627625	287.2650146	143.6355133	1
114.9644165	126.2941818	99.96736908	150.4633026	287.4805908	143.6482239	1
114.9450912	126.2749557	99.96736908	150.5478516	287.6619568	143.658905	1
114.9254456	126.2547302	99.96736908	150.6193085	287.8153381	143.6678925	1
114.8981857	126.226181	99.96736908	150.6791992	287.9438782	143.6754608	1
114.8617249	126.1866913	99.96736908	150.7298737	288.0527954	143.6817627	1
114.8220749	126.1441956	99.96736908	150.7720642	288.1433411	143.687088	1
114.7676544	126.0853348	99.96736908	150.807373	288.2191772	143.6915283	1
114.718399	126.0313339	99.96736908	150.8372955	288.2835083	143.6952667	1
114.6688614	125.9780273	99.96736908	150.8617706	288.3360596	143.698349	1
114.6068726	125.9099274	99.96736908	150.882782	288.3812561	143.7009583	1
114.5505981	125.8485184	99.96736908	150.9000244	288.4183044	143.703125	1

114.4817963	125.7734985	99.96736908	150.9143524	288.4490967	143.7049255	1
114.416687	125.7022171	99.96736908	150.9263611	288.474884	143.7064209	1
114.3512955	125.6305695	99.96736908	150.9363098	288.4962769	143.7076721	1
114.2800217	125.5525742	99.96736908	150.9445648	288.5140076	143.708725	1
114.204483	125.46978	99.96736908	150.9514771	288.5288696	143.7095642	1
114.131073	125.3890152	99.96736908	150.9572906	288.5413818	143.7102814	1
114.0570145	125.3080139	99.96736908	150.961853	288.5511475	143.7108917	1
113.9695816	125.2113342	99.96736908	150.9661713	288.5605164	143.7113647	1
113.8954926	125.1304016	99.96736908	150.9691315	288.566864	143.7117767	1
113.8126602	125.0395813	99.96736908	150.9717865	288.5725098	143.7121124	1
113.7275848	124.9460144	99.96736908	150.9740143	288.5773621	143.7123871	1
113.643692	124.8537292	99.96736908	150.9759521	288.5815125	143.712616	1
113.5568542	124.7582169	99.96736908	150.9775085	288.5848999	143.7127991	1
113.4684906	124.6612015	99.96736908	150.9787598	288.5875549	143.7129669	1
113.3714066	124.5545807	99.96736908	150.9797211	288.5896301	143.7130585	1
113.2850647	124.4600677	99.96736908	150.9803467	288.5909729	143.7131653	1
113.188797	124.3543854	99.96736908	150.9809875	288.5923157	143.7132721	1
113.0925064	124.2482758	99.96736908	150.9817657	288.5940552	143.7133331	1
112.9946594	124.1411972	99.96736908	150.9819336	288.5943298	143.7134094	1
112.9011688	124.0384445	99.96736908	150.9823914	288.5953369	143.7134399	1
112.8032532	123.9305801	99.96736908	150.9828186	288.596283	143.7134705	1
112.692276	123.8089752	99.96736908	150.9829102	288.596405	143.713501	1
112.5900421	123.696312	99.96736908	150.9833069	288.5973206	143.7135315	1
112.4800491	123.575386	99.96736908	150.9834442	288.5976257	143.7135315	1
112.3715973	123.4565659	99.96736908	150.983429	288.5975647	143.7135773	1
112.2706604	123.3458176	99.96736908	150.983429	288.5975647	143.7135773	1
112.1530457	123.2160263	99.96736908	150.9838562	288.5985413	143.7135773	1
112.0446472	123.0973282	99.96736908	150.9836731	288.5980835	143.7136078	1
111.9258041	122.9665451	99.96736908	150.9839172	288.5986633	143.7136078	1
111.8116074	122.8413849	99.96736908	150.9837646	288.5983276	143.7136078	1
111.6891861	122.7063522	99.96736908	150.9840851	288.5990601	143.7136078	1
111.5638657	122.5689468	99.96736908	150.9839783	288.5987549	143.7136078	1
111.4404984	122.4331894	99.96736908	150.9841461	288.5991516	143.7136078	1
111.3131409	122.2937698	99.96736908	150.9837952	288.5983887	143.7136078	1

111.1849899	122.1525116	99.96736908	150.9840851	288.5990295	143.7136078	1
111.0546646	122.0099487	99.96736908	150.9837494	288.5982361	143.7136078	1
110.9222412	121.8640594	99.96736908	150.9839935	288.5988159	143.7136078	1
110.7859573	121.7141418	99.96736908	150.9841461	288.5991516	143.7136078	1
110.6490631	121.5638962	99.96736908	150.9840546	288.5989685	143.7136078	1
112.4883728	122.7746582	99.96736908	160.018631	298.8968506	151.5462494	1
110.0062332	117.9115677	99.96736908	175.8639221	265.3945618	211.7306366	1
110.090683	117.7195129	99.96736908	175.7397461	263.2410583	213.8247223	1
110.2076492	117.5625534	97.53217316	176.924881	261.2940674	215.7749634	1
110.3573151	117.4393539	94.33287811	178.5485229	259.5371094	217.5927277	1
110.5453873	117.3575974	91.19776917	180.1712036	257.9519653	219.2893066	1
110.7640228	117.3095474	88.13082123	181.7861786	256.522644	220.8726501	1
111.0192795	117.3028183	85.14216614	183.3838196	255.2341614	222.3506927	1
111.2980194	117.3243408	82.23130035	184.9602051	254.0727692	223.7302246	1
111.6047897	117.3797836	79.40706635	186.5069733	253.0261841	225.0173492	1
111.9392776	117.4696655	76.67493439	188.0176239	252.0829163	226.2175751	1
112.2962952	117.5891647	74.0364151	189.4886169	251.2328339	227.3365173	1
112.6633072	117.7258148	71.49202728	190.9167633	250.4669037	228.3783875	1
113.0573959	117.8979645	69.0519104	192.2944489	249.7764282	229.3479309	1
113.4650497	118.0920563	66.71362305	193.6210175	249.1540375	230.2493134	1
113.8901138	118.3124237	64.48098755	194.8927917	248.5929565	231.0864868	1
114.3227081	118.549118	62.35167694	196.1096802	248.0869751	231.8633575	1
114.7608948	118.8001175	60.32703781	197.2697296	247.6306763	232.5832062	1
115.2055435	119.0670624	58.40558243	198.3732758	247.2190857	233.2503815	1
115.6459503	119.3383942	56.58599091	199.4198151	246.8476563	233.867218	1
116.0949631	119.6277695	54.86974716	200.4083405	246.5123749	234.4373474	1
116.5347595	119.9165726	53.25126648	201.3414001	246.2097778	234.9636383	1
116.9664383	120.205658	51.72955704	202.2191315	245.9364777	235.4487305	1
117.3964767	120.5020599	50.30197906	203.0431671	245.689621	235.8960876	1
117.8158264	120.7957382	48.96476746	203.8152771	245.4665985	236.3079376	1
118.2175903	121.0801392	47.71337509	204.5377197	245.2649231	236.6866913	1
118.6197968	121.3727036	46.54831696	205.2105865	245.0826874	237.0348053	1
118.9972153	121.6476212	45.46117783	205.8381958	244.9178314	237.3544159	1
119.3676147	121.923027	44.45162201	206.4210052	244.7687073	237.6476593	1

119.7080612	122.1746902	43.51232529	206.9629822	244.6338196	237.9164886	1
120.0441589	122.4288559	42.64321899	207.464386	244.511673	238.1627655	1
120.3684769	122.6721573	41.84552383	207.9260712	244.4034729	238.3882141	1
120.6620178	122.8944778	41.10194778	208.3549194	244.3037109	238.5944519	1
120.9451523	123.1102448	40.41848755	208.7495422	244.2142029	238.78302	1
121.2093582	123.3164749	39.78372192	209.114624	244.1312866	238.9552307	1
121.4598312	123.5124512	39.20087051	209.4500732	244.0566864	239.1125031	1
121.6904526	123.6943054	38.66318512	209.7590332	243.9886017	239.256012	1
121.9056854	123.8620529	38.17256546	210.0416107	243.9280396	239.3868256	1
122.1006088	124.0183945	37.71608734	210.3028564	243.8710022	239.5060425	1
122.2838745	124.164444	37.30047226	210.5412292	243.8202209	239.6145935	1
122.4579315	124.3054657	36.91954803	210.7594299	243.7738342	239.7133942	1
122.6125107	124.4288635	36.57154465	210.9588928	243.732193	239.8032837	1
122.762886	124.5483093	36.2575531	211.1396179	243.6955719	239.8849792	1
122.8937912	124.6545181	35.96638107	211.3060608	243.6609802	239.9591675	1
123.0109558	124.7547913	35.69358826	211.4602203	243.6271057	240.026535	1
123.122406	124.8457336	35.45205688	211.5982361	243.5988312	240.0876465	1
123.2242355	124.9288864	35.23216248	211.7238464	243.5731964	240.1430817	1
123.3169403	125.0052567	35.0317955	211.8381195	243.5498352	240.1932678	1
123.3976059	125.0747604	34.84482574	211.943573	243.5270386	240.238739	1
123.4810486	125.1410751	34.68474579	212.0359802	243.5095825	240.2798767	1
123.5535889	125.2000809	34.53686905	212.1206512	243.4928894	240.3170624	1
123.6142273	125.2511368	34.39923477	212.1986389	243.4767151	240.350708	1
123.6142273	125.2511368	34.39923477	212.1986389	243.4767151	240.350708	1